

SPMM
Smart
Revise

Epidemiology

Paper B

Syllabic content 14

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1. Introduction

- ★ Psychiatric epidemiology deals with counting the frequency, risk factors and burden of psychiatric diseases in specified human population in a specified period. The essence of epidemiology is studying the distribution of disease using which causal attributions could be drawn and health services could be organised.

- ★ **Descriptive epidemiology** deals with describing the pattern of diseases while **analytical epidemiology** attempts to make

Triad of epidemiological questions	
TIME	How does the pattern of disease vary over time?
PLACE	How does the place in which the population lives affect the disease pattern?
PERSON	How do the personal characteristics of the people in a population affect the disease pattern?

associations of observed patterns in order to make inferences in a hypothesis driven manner.

- ★ A fast expanding branch of psychiatric epidemiology is **genetic epidemiology**. It is the epidemiological evaluation of the role of inherited causes of disease in families and in populations; Gene-gene and gene-environment interactions are also studied in genetic epidemiology. Morton defined genetic epidemiology as 'a science which deals with the etiology, distribution, and control of disease in groups of relatives and with inherited causes of disease in populations.'

2. Key Terms in Epidemiology

The incidence of a disease is defined as the number of 'new' cases diagnosed in a specified time interval for a specified size of the population at risk. Mid-interval population usually determines this population size. For example, if the incidence of a disease in one year is calculated, then the comparison will be made against the mid-year population. To be accurate, the comparison must exclude those who are not risk, but this is generally not done for non-infectious, non-epidemic diseases such as psychiatric illnesses.

The essential criterion is that the measure should indicate all new occurrences of a disease within the period of observation in an area, irrespective of whether the newly diagnosed patients are cured or dead well within the period of observation itself.

$$\text{Incidence} = \frac{\text{Number of newly diagnosed cases}}{\text{-----}}$$

(In the year 2014) Mid 2014 population at an area

Incidence is a ratio measured against time. One-month incidence will be invariably lower than one-year incidence. The most common measurement of incidence using the above formula is also called as **cumulative incidence**. This is valid in cohort studies where attrition is low, but not so useful when attrition is significant. This is because dropouts are excluded from being counted as new cases, while the denominator population usually takes them into account. To avoid this problem, incidence density (also called as **incidence rate**) is used. It expresses number of new cases per person-year of observation. The unit of the incidence rate is rather bizarre, time⁻¹. Hence, incidence rate ratios are often used in reporting results that compare two groups' incidence rates.

Prevalence is defined as the number of 'existing' cases in a specified population for a period of observation (either cross-sectional observation – called point prevalence, or longitudinal observation for a specified time – called period prevalence). The existing cases include all new cases, all cases diagnosed before the observation but still suffering from the disease. But existing cases exclude those who have been previously diagnosed but now cured or dead. The essential criterion is that the cases must be found when observation is being carried out, irrespective of when the diagnosis occurred. This is not possible if someone is dead or cured well before the observation period.

Number of existing (both [new + old] – [dead + cured])

Point prevalence = -----

(On Jan 28, 2008) Cross sectional population at an area on Jan 28, 2008

Number of existing (both [new + old] – [dead + cured])

Period prevalence = -----

(2007 - 2008) Mid year population at an area

For illnesses that are significantly chronic (e.g. schizophrenia), prevalence will be higher compared to those illnesses that are acute and short-lived (e.g. influenza), even if the incidence rates are comparable. Hence a simple expression below explains the relationship between incidence and prevalence:

Prevalence = incidence X duration of illness

Certain factors can influence incidence and prevalence differently. For example, if a new vaccine is developed for preventing illness, both incidence and prevalence may come down. If a cure is developed for schizophrenia, the incidence may not be affected, but prevalence can drop. Similarly, if interventions are introduced to reduce mortality in chronic schizophrenia, then prevalence may paradoxically increase due to the longevity of patients. This may not affect incidence rates directly.

Lifetime prevalence is the proportion of individuals in the population who have ever manifested a disorder, who are alive on a given day. This is ascertained by surveying a population cross-sectionally and

finding out if they ever satisfied the criteria for a disorder in the past or currently. As one can observe, this method though commonly used in epidemiological surveys, is prone to recall bias.

Lifetime prevalence needs to be clearly distinguished from *lifetime morbid risk* (LMR). LMR is the probability of a person developing the disorder during the entire period (often a specified period, defined by the life expectancy of the population studied) of their life. LMR includes the entire lifetime of a birth cohort both past and future and includes those deceased at the time of the survey. For low-incidence disorders such as schizophrenia, summation of age-specific incidence rates gives approximate LMR (Saha et al., 2004)

Epidemic, endemic and pandemic

- ★ The usual prevalence of a disease in a community is called the **baseline** prevalence.
- ★ If the baseline is low to a moderate level and continues without significant fluctuation, the disease is said to be **endemic** in the community.
- ★ A persistent but high level of the baseline is called **hyperendemic** occurrence. Most psychiatric disorders are low-level endemic disorders (with exceptions).
- ★ An irregular pattern of significant fluctuation from baseline is called **sporadic** occurrence.
- ★ When the occurrence within an area clearly exceeds the expected level in a given time period, it is called an **epidemic**. An epidemic can be slow and steady rising epidemic as in the case of multifactorial chronic diseases or acute outbreaks as in infectious diseases.
- ★ A **pandemic** is an epidemic in wide geographical proportion e.g. HIV pandemic.

Mortality rates

Crude, specific and standardized rates: A crude rate is any rate applicable to whole population e.g. crude birth rate, crude mortality rate, etc. A specific rate is any rate applicable to a subgroup of a population e.g. age-specific mortality, disease-specific mortality, cause-specific mortality rates, etc. A standardized rate is a rate applicable to a hypothetical population with an adjusted variable e.g. age. As population samples are heterogeneous, crude rates from one population may not be comparable to another population. For example suicide rates in inner London may not be comparable to rates in rural Yorkshire, as working age population may be more in London, spuriously increasing suicide rates. Hence, standardized hypothetical populations are used on which observed rates from a population are applied, and adjusted values are derived. These standardized values are easily comparable.

Crude mortality rate: Ratio between number of deaths due to all-cause in a population and total population size

Cause-specific mortality rate: e.g. alcohol specific mortality. This refers to the ratio between number of deaths due to alcohol in a population and total population size.

Case-fatality rate: Ratio between number of deaths due to a specific disease and number of persons affected by the disease in a population. It is a measure of the fatal severity of the disease studied. For example, 15 patients out of 100 with anorexia will die due to its complications.

Proportionate mortality rate: It is a measure of the contribution of a disease to the societal mortality burden. It is given by the ratio between deaths due to a specific cause and total number of deaths in a population.

Years of Potential Life Lost (YPLL) is a measure of the impact of premature mortality on a population. It is calculated as the sum of the differences between some predetermined end point and the ages of death for those who died before that end point. The two most commonly used end points are age 65 years and average life expectancy. Because of the way in which YPLL is calculated, this measure gives more weight to an earlier deaths than later deaths.

Disability-adjusted life years - The DALY is a health gap measure that extends the concept of potential years of life lost due to premature death (PYLL) to include equivalent years of 'healthy' life lost in states of less than full health, broadly termed disability. One DALY represents the loss of one year of equivalent full health. Additionally, 3% time discounting and non-uniform age weights that give less weight to years lived at young and older ages were used in calculating DALYs.

DALYs for a particular disease is given by

$$\text{DALY} = \text{YLL} + \text{YLD}$$

YLL - years of life lost due to premature mortality in the population

YLD - years lost due to disability for cases.

$$\text{YLL} = N \times \text{LE}$$

- Where N = number of deaths and
- LE = standard life expectancy at age of death in years

$$\text{YLD} = I \times \text{DW} \times \text{LD}$$

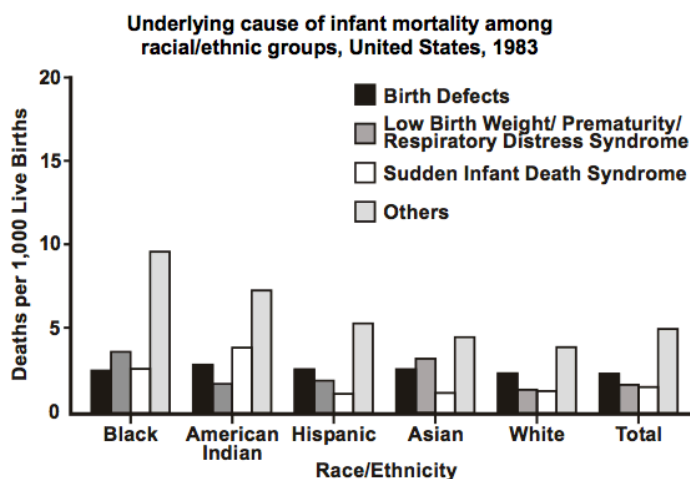
- I = number of incident cases
- DW = weight given to the disability
- LD = average duration of the case until remission or death (life until death in years)

Neonatal death	• A liveborn that dies within 28 days
Early neonatal death	• A liveborn that dies within 7 days
Late neonatal death	• A liveborn that dies between 7 and 28 days
Stillbirth	• A fetus that dies before birth but after 24-weeks of gestation
Perinatal death	• still births + early neonatal biths
Postneonatal deaths	• deaths from 1 month to 1 year
Infant deaths	• deaths under 1 year age

Note that for most of the above rates, total number of live births is the denominator.

3. Graphical Expressions in Epidemiology

Bar charts are used to display the data where each category of a variable is represented by a bar. The



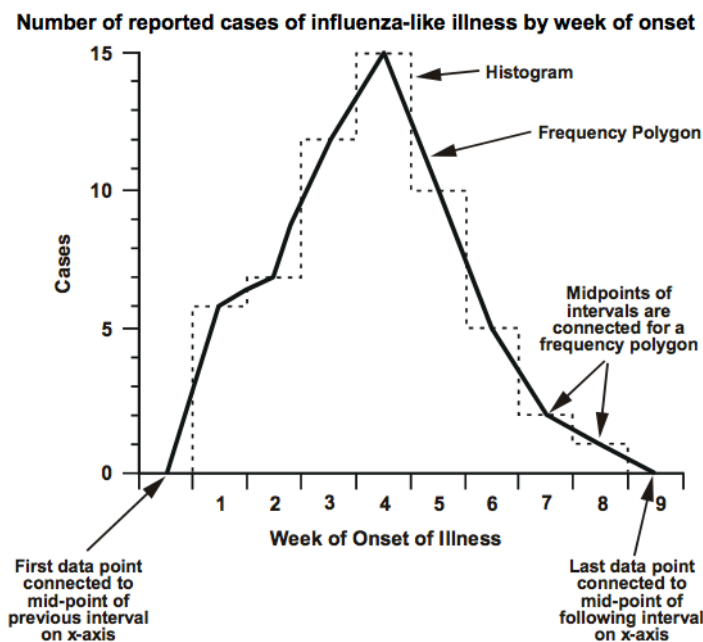
length of the bar is proportional to the number of persons or events in that category.

A **frequency polygon** is the graph of a frequency distribution. In a frequency polygon, we mark the number of observations within an interval with a single point placed at the midpoint of the interval and then connect each set of points with a straight line.

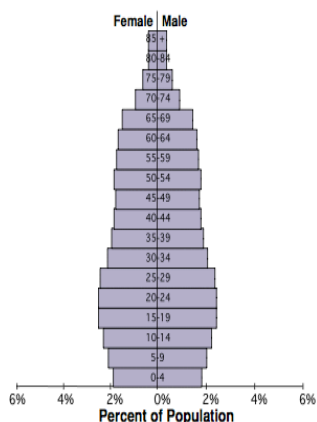
(From Centers for Disease Control and Prevention (CDC)
Training and Continuing Education Online)

A **population pyramid** is a useful tool for understanding the structure and composition of populations because it graphically portrays many aspects of a population, such as sex ratios and age structure. Pyramids can give insight into trends in population over time by their portrayal of the relative number of people in a particular cohort.

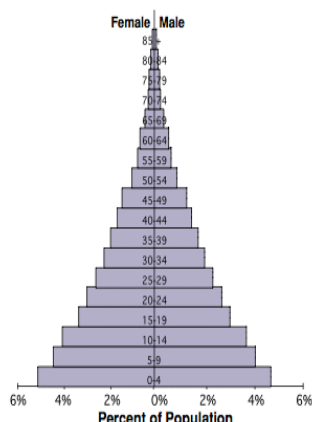
Demographers recognize three prototypical pyramid structures, constrictive, expansive, and stationary.



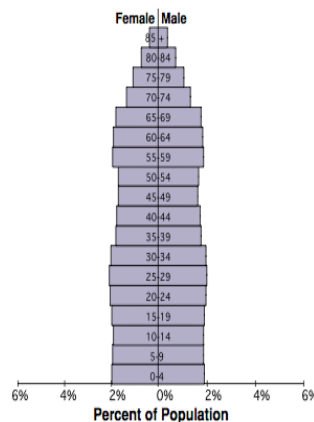
1. A **constrictive pyramid** has fewer people in the younger age categories and has been typical of U.S. populations as baby boom populations shift to more conservative birth rates.
2. An **expansive pyramid** represents greater numbers of people in the younger age categories and is typical of many developing countries where birth rates are high, but conditions are harsh, and life expectancy is short.
3. A **stationary pyramid** shows roughly equal numbers of people in all age categories, with a tapering towards the older age categories. Countries such as Sweden show stationary age



Constrictive



Expansive



Stationary

categories because of relatively low, constant birth rates, and a high quality of life

4. History of Psychiatric epidemiology

To describe the development of psychiatric epidemiology, three 'generations' of studies are distinguished. Around 16 psychiatric-epidemiological studies carried out before World War 2 belongs to the **first**

generation (includes Midtown Manhattan study). These studies focused primarily on the health care agency registered prevalence of mental disorders in relation to community characteristics. The **second generation** of psychiatric epidemiological studies followed an increased interest in the diagnostic criteria, classification and nomenclature of psychiatric disorders after World War 2 where nearly 60 studies appeared. These were largely field surveys conducted in unstructured clinical interviews. Consequently, the reliability of these studies was low. The **third generation** studies started around 1970 with more effort into increasing the reliability of psychiatric diagnoses. A major objective of 3rd generation studies is to obtain precise estimates of prevalence and incidence of specific mental disorders, whereas 2nd generation studies focused on mental ill-health in general. It is claimed that a **fourth generation** of psychiatric epidemiological studies is in the making. This includes studies which include comprehensive sets of biologic markers including brain imaging, cerebrospinal fluid examinations, blood sampling, etc. in the large scale cross-sectional surveys e.g. H70 study (Skoog 2004).

DISCLAIMER: This material is developed from various revision notes assembled while preparing for MRCPsych exams. The content is periodically updated with excerpts from various published sources including peer-reviewed journals, websites, patient information leaflets and books. These sources are cited and acknowledged wherever possible; due to the structure of this material, acknowledgements have not been possible for every passage/fact that is common knowledge in psychiatry. We do not check the accuracy of drug-related information using external sources; no part of these notes should be used as prescribing information

References and further reading

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- ★ Maj & Sartorius, WPA series Evidence and experience in Psychiatry: Schizophrenia pp 249
- ★ Compton MT. Clinical Manual of Prevention in Mental Health 2010, American Psychiatric Press